Jedidiah Alindogan

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Education

California Institute of Technology (Caltech)
Ph.D. in Control and Dynamical Systems, in progress — Sep 2024 - Present
B.S. in Mechanical Engineering — Sep 2021 - Jun 2023

Research Experience

Graduate Student, Caltech — Sep 2024 - Present
Reinforcement learning for snake-robot locomotion; Caltech–NASA JPL

- Developed a planar locomotion controller for an 18-DOF snake robot using Proximal Policy Optimization (PPO) in Python and MuJoCo.
- Used sinusoidal gait parameters as policy actions to track commanded linear and angular velocities.
- Designed a pose-tracking reward by integrating velocity commands with a unicycle model to address oscillatory velocity feedback.
- Validated zero-shot sim-to-real transfer on concrete and gravel during building-foundation inspection trials using policies trained in MuJoCo.

Research Engineer, Caltech — Aug 2023 - Aug 2024
Autonomous Robotics and Control Lab

- Built a robotics payload for a Traxxas X-Maxx to run planning and control algorithms; designed the enclosure in SolidWorks and integrated sensors and motor interfaces with a Jetson Orin.
- Calibrated motors using ODrive and configured sensor calibration and odometry tools for the vehicle platform.
- Developed a physics-based tipping constraint that accounts for turning forces on slopes.
- Built Gazebo test terrains and evaluated neural-network controllers on hardware across different terrain conditions.

Technical Skills

Programming: Python, C++, C#, MATLAB, Mathematica
Robotics and learning: PyTorch, MuJoCo, PPO, ROS 1, ROS 2, Gazebo
Engineering tools: Git, Arduino, SolidWorks, Fusion 360, ANSYS

Publications

Vertical Mobility and RL-Based Locomotion of a Snake-Like Robot for Building Foundation Inspection. Coauthor. IEEE/RSJ IROS, 2026.

MAGIC-VFM: Meta-learning Adaptation for Ground Interaction Control with Visual Foundation Models. Coauthor. IEEE Transactions on Robotics, 2025.

Model Predictive Trees: Sample-Efficient Receding Horizon Planning with Reusable Tree Search. Coauthor. IEEE/RSJ IROS, 2024.

Leadership and Honors

Academics Chair, Caltech Graduate Student Council — 2024 - Present
Organize academic and career activities for graduate students.
Tau Beta Pi, member — 2022